

Quiz: Dynamic Memory and Structs in C

Total: 10 points

Question: Dynamic Struct Management (10 points total)

You will design, use, and free a dynamically allocated struct in C that represents a **Book**. The struct will contain one pointer field for a string.

(a) (3 pts) Define a struct named **Book** that contains:

- `char *title;` — a pointer to a dynamically allocated string
- `int pages;` — the number of pages
- `float price;` — the price of the book

Write your struct definition below:

```
typedef struct {
```

```
} Book;
```

(b) (3 pts) Write a function that takes a pointer to a **Book** and frees all associated memory using the following header:

```
void free_book(Book *b);
```

Your function should:

- Free the memory allocated for the title
- Free the struct itself

Write your function below:

```
void free_book(Book *b) {
```

(c) (4 pts) Write a `main()` function that:

- Dynamically allocates memory for one `Book` struct
- Allocates and assigns memory for its `title`
- Assigns values to `pages` and `price`
- Prints its contents as:

Title: <title>, Pages: <pages>, Price: \$<price>
- Calls `free-book()` to clean up allocated memory

```
int main(void) {
```

End of Quiz.